GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

COMBUSTION MONITOR

CARD APPLICATION NOTES

GEK 65322

GENERAL ELECTRIC

COMBUSTION MONITOR

CARD APPLICATION NOTES

CONTENTS

SECTION 1: CARD APPLICATION NOTES

1.0.0 INTRODUCTION

1.1.0 GENERAL FUNCTION

1.2.0 HARDWARE CONFIGURATION

1.2.1 ANALOG INPUT SCANNING SYSTEM

1.2.1-1 NCJA - COLD JUNCTION COMPENSATION BRIDGE CARD

1.2.1-2 NITA - ISOTHERMAL TERMINATION CARD

1.2.1-3 NTSA - THERMOCOUPLE SCANNER CARD

1.2.1-4 NADC - ANALOG INPUT CARD

1.2.2 CENTRAL PROCESSING UNIT

1.2.2-1 HMPA - MICROPROCESSOR CARD

1.2.3 INPUT/OUTPUT INTERFACE (WITH TURBINE CONTROL PANEL)

1.2.3-1 HLIC ISOLATED INPUT CARD

1.2.3-2 HRRA RELAY OUTPUT CARD

1.2.4 OPERATOR DISPLAY MODULE

1.2.4-1

1.2.5 POWER SUPPLY

REV. 1 1/25/79 A C	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC
REV. 2	ISSUED 9-15-78			FIRST MADE FOR	
REV. 3	MADE BY AZRA EHSAN			I.C. NO.	

ELEMENTARY DIAGRAM
COMBUSTION MONITOR
CARD APPLICATION NOTES

GEK 65322

FORM 1C 2390 4-74

ORDER I NG- FIXED DES IGN

SELECT FORM FROM TABLE

DS3840CM1

FORM ——— V I NTAGE (FOR DEPARTMENT I NTERNAL CONTROL USE ONLY)

FORM	LEFT MTG	RIGHT MTG	NTSA (QTY)	WITH HXPA	WITHOUT HXPA	WATER TRIP	NO WATER TRIP
AA	X		2		X	X	
AB	X		2			X	
AC	X		2		X		X
AD	X		2				X
AE	X		3		X	X	
AF	X		3			X	
AG	X		3		X		X
AH	X		3				X
AX	X		2				
BA		X	2		X	X	
BB		X	2			X	
BC		X	2		X		X
BD		X	2				X
BE		X	3		X	X	
BF		X	3			X	
BG		X	3		X		X
BH		X	3				X
BX		X	2				

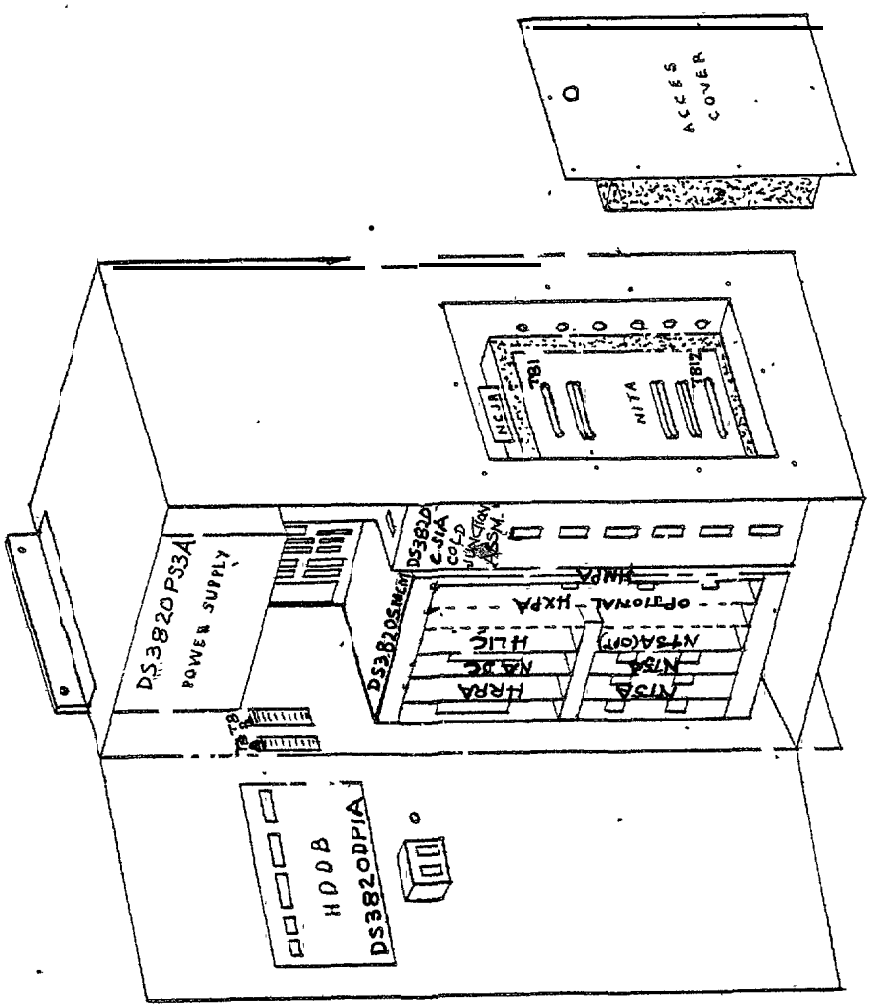
DETA I LS REQU I RED

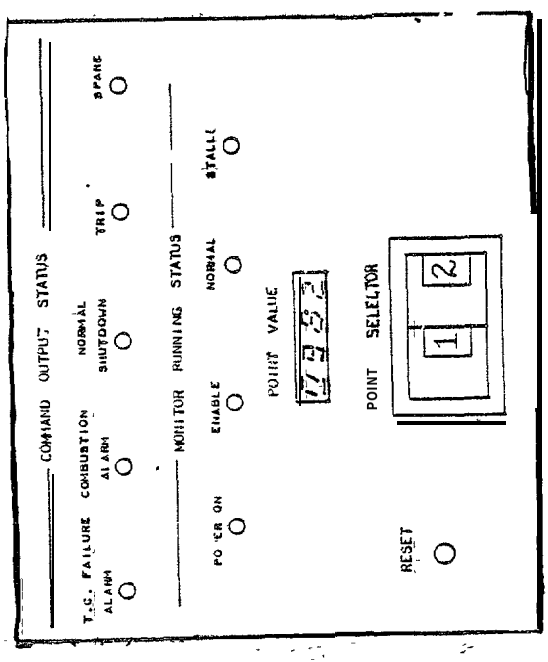
1. TC TYPE-CHROMEL-ALUMEL/IRON-CONSTANTAN.
2. TELETYPE® TERMINAL BAUD RATE 110 BAUD (10 CHARACTERS/SEC.) 300 BAUD. (30 CHARACTERS/SEC.)
3. TEMPERATURE DISPLAY °F/°C.

EXPANSION CAPABILITY FOR FUTURE IMPLEMENTATION.

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FORM 1C 2390 12-74





DISPLAY PANEL LAYOUT

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ELEMENTARY DIAGRAM CARD
APPLICATION NOTES
GEK 65322
CONT. ON SH. 4 SH. NO. 3

COMBUSTION MONITOR

1.0.0 INTRODUCTION: THE COMBUSTION MONITOR DS3840CM1 IS MICROPROCESSOR BASED SYSTEM DESIGNED TO MONITOR THE COMBUSTION SYSTEM OF A GAS TURBINE, THE DESIGN IS MODULAR BY FUNCTION CONSISTING OF (SEE FIG.!).

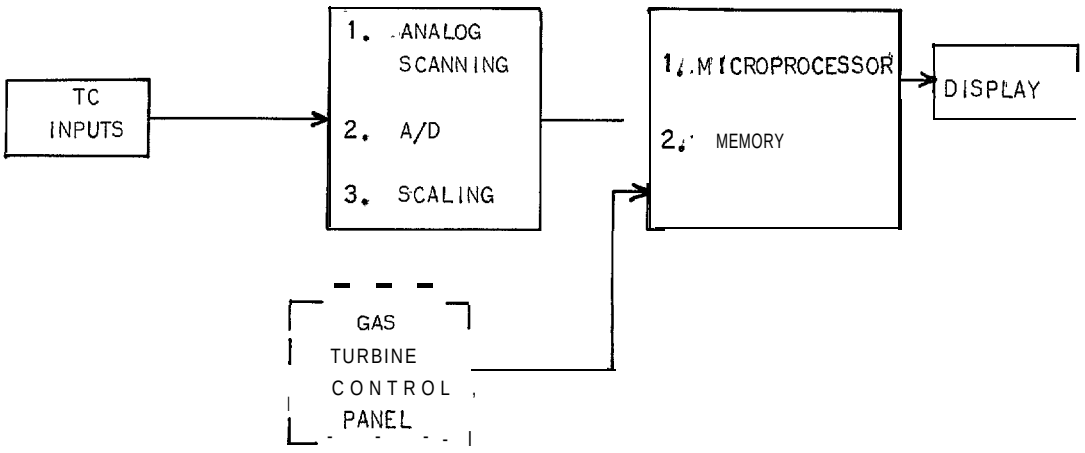
1. AN OPERATOR DISPLAY PANEL, DS3820DP1
2. A SMALL ELECTRONICS MODULE (SEM), CONTAINING THE MICROPROCESSOR BOARD, THE DIGITAL INPUT AND OUTPUT BOARDS, AND THE ANALOG SCANNER BOARDS, DS3820SMC
3. A POWER SUPPLY MODULE.. DS3820PS3
4. AN ISOTHERMAL TERMINATION BOX,. DS3820CJ1
5. AN ENCLOSURE FOR THESE MODULES,.

1.1.0 GENERAL FUNCTION

THE COMBUSTION MONITOR EXAMINES THE TURBINE EXHAUST TEMPERATURE THERMOCOUPLES AND THE COMPRESSOR DISCHARGE TEMPERATURE THERMOCOUPLES TO DETECT TEMPERATURE PATTERNS WHICH INDICATE POSSIBLE COMBUSTION SYSTEM PROBLEMS. FROM CHANGES THAT OCCUR IN THE TEMPERATURE PATTERN, WARNING SIGNALS ARE GENERATED BY THE MONITOR AND SENT TO THE GAS TURBINE CONTROL PANEL, (SEE FIG, 2)

REV. 1 1/25/79 AE	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM COMBUSTION MONITOR CARD APPLICATION NOTES
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REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH. 5 SH. NO. 4

1.2.0 HARDWARE CONFIGURATION:



THE COMBUSTION MONITOR HARDWARE CONFIGURATION MAY BE SEGMENTED INTO THE FOLLOWING MAJOR FUNCTIONAL BLOCKS:

I. ANALOG SCANNING SYSTEM CONSISTING OF THE FOLLOWING CARDS:

ISOTHERMAL BOX
1. DS3800 NCJA - RTD COLD JUNCTION COMPENSATION BRIDGE
2. DS3800 NITA - ISOTHERMAL TERMINATION CARD
3. DS3800 NTSA - THERMOCOUPLE SCANNER CARDS (2)
4. DS3800 NADC - ANALOG INPUT CARD

II. CENTRAL PROCESSING UNIT CONSISTING OF THE FOLLOWING CARD:

1. DS3800 HMPA - MICROPROCESSOR CARD

III. INPUT/OUTPUT INTERFACE TO GAS TURBINE CONTROL PANEL CONSISTING OF THE FOLLOWING CARDS:

1. DS3800 HLIC - ISOLATED LOGIC -INPUT CARD
2. DS3800 HRRA - RELAY OUTPUT CARD

IV. OPERATOR DISPLAY MODULE: CONSISTING OF THE FOLLOWING CARD

1. DS3800 HDDB - DISPLAY INTERFACE CARD

V. POWER SUPPLY CONSISTING OF:

1. DS3820 PS3A

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REV. 3	MADE BY AZRA EHSAN		I.C. NO		CONT. ON SH. 6 SH. NO. 5

FORM 244A9738P1 (9-20-71)

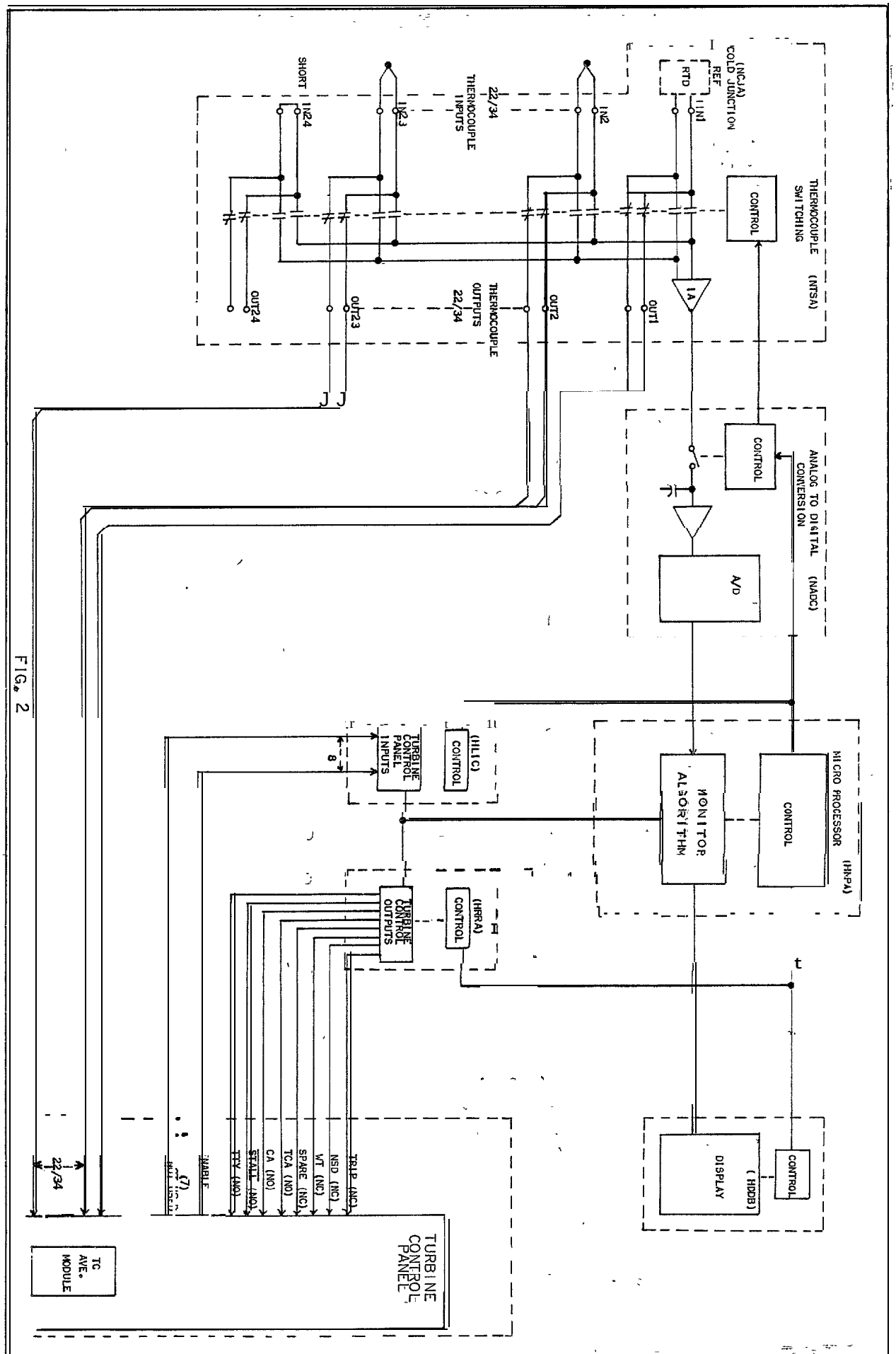


FIG. 2

REV. 3	REV 2	REV 1 1/25/79	APPROVAL Azra Ehsan	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM CARD APPLICATION NOTES
REV 4	ISSUED 9-15-78	PRINTS TO	FIRST MADE FOR REL		GEK 65322 SH NO 6
REV 5	MADE BY AZRA EHSAN		I C NO		CONT ON SH 7

FORM 1C 2390 12-74

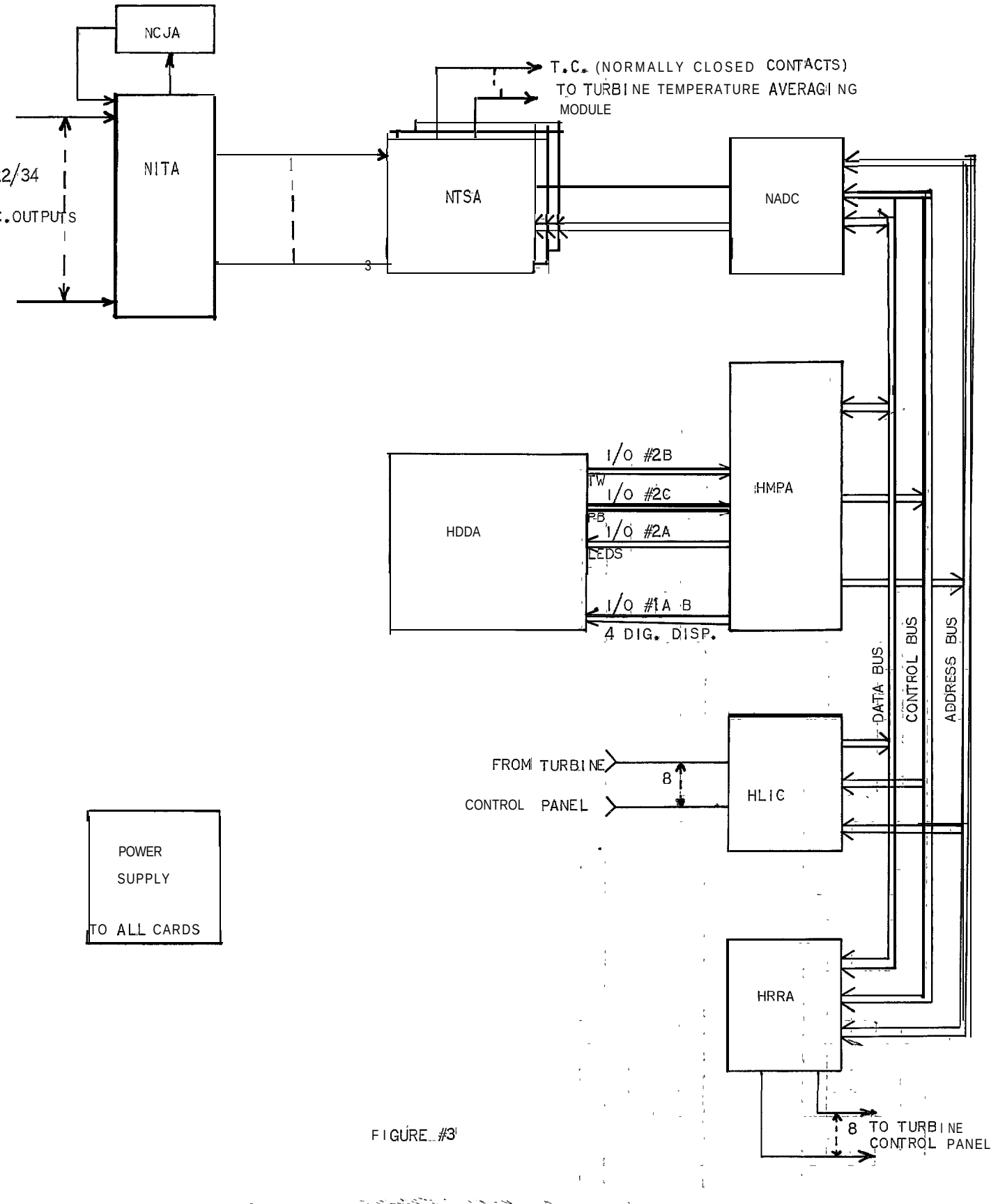


FIGURE #3

REV. 1 1/25/79 AE	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC
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ELEMENTARY DIAGRAM
COMBUSTION MONITOR
CARD APPLICATION NOTES

GEK 6 5 3 2 2

CONT ON SH. 8 SH. NO 7

A B C D E F G H J K L

1.2.1 ANALOG INPUT SCANNING SYSTEM

Up TO 34 CHROMEL-ALUMEL OR [RON-CONSTANTAN THERMOCOUPLES CAN BE SCANNED By THE PROCESSOR,

THE TURB I NE EXHAUST TEMPERATURE THERMOCOUPLES AND COMPRESSOR DISCHARGE TEMPERATURE THERMOCOUPLES CONNECT TO THE N ITA TERM INAL BOARDS INS IDE THE ISOTHERMAL BOX DS3820CJ1. THESE THERMOCOUPLE s IGNALS CONNECT THROUGH NORMALLY CLOSED CONTACTS ON DUAL FORM - C MERCURY WETTED RELAYS ON THE NTSA CARDS BACK TO TERMINALS ON THE NITA CARD, FROM THESE TERMINALS THE THERMOCOUPLES CAN BE W I RED TO THE TURBINE CONTROL PANEL, THE THERMOCOUPLE S IGNALS CONNECT THROUGH THE NORMALLY OPEN CONTACTS OF THE RELAYS TO THE NTSA CARDS; THE PROCESSOR COMMUNICATES DIRECTLY WITH A MEMORY MAPPED IO PORT ON THE NADC CARD To ENERGIZE ONE NTSA RELAY AT A TIME. WHEN A, RELAY IS ENERGIZED THE SELECTED THERMOCOUPLE S IGNAL IS CONNECTED TO A PRECISION INSTRUMENTATION AMPLIFIER ON THE NTSA, THE AMPLIFIED THERMOCOUPLE S IGNAL THEN GOES TO AN A/D CONVERTER WHICH CAN BE ADDRESSED BY THE PROCESSOR. A SIGNAL MEASURING THE COLD JUNCTI ON TERMINATION TEMPERATURE IS AVA I LABLE ON THE FIRST ADDRESSABLE NTSA RELAY,

1.2.1-1 NCJA COLD JUNCT ION COMPENSATION BRIDGE CARD

THE DS3800 NCJA CARD CONS ISTS OF A RES ISTIVE BRI DGE (RTD) DESIGNED FOR MEASURING THE "COLD JUNCTION" TEMPERATURE (TCJ) . THE RTD IS PHYS ICALLY MOUNTED ON THE ISOTHERMAL CARD (N I TA) .

THE NCJA CARD IS CALIBRATED AT THE FACTORY AND SHOULD NOT REQUIRE CALIBRATION DURING THE FIRST SIX MONTHS OF OPERATION, SEE SPECIFICATION CONTROL - COMBUSTION MONITOR (287A1444) FOR CALIBRATION PROCEDURE, RECALI BRAT ION IS NOT NECESSARY FOR PRON-CONSTANTAN THERMOCOUPLES.

THE TCJ VALUE IS READ ONCE EACH SCAN CYCLE BY THE CPU, THE COLD JUNCTION COMPENSAT ION IS MADE BY THE SYSTEMS SOFTWARE, THE COLD JUNCTION VALUE READ AT THE BRIDGE OUTPUT IS FOUR TIMES THE CHROMEL ALUMEL JUNCT [ON AT THAT TEMPERATURE,

COLD JUNCT ION BR I DGE OUTPUT GRADIENT : 163 uV/° C

COLD JUNCTION OUTPUT AT 25" C : 4.0 MV

1.2.1-2 NITA - ISOTHERMAL TERM I NAT I ON CARD

THIS [s THE INTERCONNECT MODULE WITH PROVISIONS FOR 34 THERMOCOUPLE INPUTS AND OUTPUTS. AN ARRAY OF TERMINAL BLOCKS IS USED FOR THE THERMOCOUPLE TERMINATIONS. k IBBON CABLE CONNECTORS FEED THE S IGNALS IN AND OUT OF THE THERMOCOUPLE SCANNER CARDS (NTSA) . ALSO MOUNTED ON THIS CARD IS THE RTD FOR THE COLD JUNCT [ON COMPENSAT I ON.

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A SHORTING BAR IS PERMANENTLY' PLACED BETWEEN TB7-P12 AND TB7-N12 FOR OFFSET COMPENSATION OF THE THERMOCOUPLE SCANNER CARD (NTSA) AMPLIF IER AND TO PREVENT THE AMPLIFIER FROM SATURATION DURING THE SCAN (THE' RELAY DROPOUT TIME IS SLOWER (3.8 M IL SEC.) COMPARED TO THE RELAY' PICK UP TIME (2 MI L. SEC.)) .

FOR A RANGE OF 32" F TO 1230°F THE MILLIVOLT S IGNAL FROM'THE THERMOCOUPLE COULD BE:

O-27.679 MV FOR CHROMEL ALUMEL T.C

O-37.009 MV FOR IRON CONSTANTAN T.C.

- MAX IMUM THERMOCOUPLE LENGTH 500 FT;
- THERMOCOUPLE WIRES MUST BE TWISTED AND ISOLATED.
- THERMOCOUPLE WIRES SHOULD BE KEPT AWAY FROM ALL OTHER W IRING.

1.2.1-3 NTSA THERMOCOUPLE SCANNER CARD

THERMOCOUPLE INPUTS ON THE NTSA CARDS ARE SCANNED ABOUT ONCE EACH SECOND BY THE CPU. EACH NTSA CARD HAS CAPABILITY OF SWITCHING TWELVE INPUT SIGNALS, MERCURY WETTED RELAYS ARE USED FOR SW ITCHING, THE COMBUST ION MONITOR HAS PROY IS IONS FOR THREE NTSA CARDS,,

THERMOCOUPLE INPUTS ON THE NTSA CARDS ARE ADDRESSED BY AN EIGHT BIT OUTPUT PORT ON THE NADC CARD. THE ADDRESSED THERMOCOUPLE INPUT RELAY IS ENERGIZED TO ISOLATE THE SIGNAL FROM THE TURBINE CONTROL PANEL AND FEED THE MILLIVOLT SIGNAL TO THE AMPLIFIER ON THE NTSA CARD. OF THE 2/3 NTSA CARDS USED IN THE SYSTEM THE AMPLIFIER ON ONLY ONE OF THE NTSA CARDS is USED TO AMPLIFY THE MILLIVOLT SIGNALS FROM ALL THE NTSA CARDS. THIS AIDS IN MINIMIZING ERRORS DUE TO DIFFERENT OFFSET OR GA IN, THE S IGNALS FROM THE AMPLIFIER OUTPUT ARE INPUT TO THE ANALOG TO DIGITAL CONVERTER ON THE NADC CARD. RELAY 1 INPUT (ON THE FIRST NCJA CARD) IS ALWAYS TIED TO THE- COLD JUNCTION MONITOR (NCJA) , RELAY 24 IS ALWAYS A SHORT.

OFFSET AND GA IN CAL IBRAT ION MAY BE NECESSARY AFTER S IX MONTHS OF OPERATION. REFER TO SPEC, CONTROL-COMBUSTION MONITOR (287A1444) FOR CALIBRATION PRO-CEDURE. CAL IBRATIONS MADE IN THE FACTORY ARE FOR CHROMEL ALUMEL THERMOCOUPLES. TO RECALIBRATE THE AMPLIFIER FOR I RON CONSTANTAN THERMOCOUPLES A BERG JUMPER ON NTSA CARD IS MOVED-FROM CH-AL AND A TO JR-CO AND A PINS, THE FINE ADJUSTMENT IS THEN MADE ON THE GAIN POTENTIOMETER R1 AS FOR THE CHROMEL ALUMEL TYPE, TEST PO I NTS TP1 AND TP2 FOR THE OUTPUT OF THE AMPLIFIER ARE AVAILABLE AT THE FRONT OF THE CARD.

- GA IN FOR CHROMEL ALUMEL INPUT 361.3
- GA IN FOR IRON CONSTANTAN INPUT 270.2

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COMMON MODE INPUT FOR THE AMPLIFIER:
MAXIMUM COMMON MODE OPERATING VOLTAGE ±10 V ,
MAXIMUM COMMON MODE VOLTAGE WITHOUT AMPLIFIER DAMAGE ±28 V

WILL WITHSTAND 1000 VDC FOR 400 USEC WITHOUT DAMAGE,

COMMON MODE REJECTION RATIO:

DC	90 DB MINIMUM	100 DB TYPICAL
AC (60 HZ)	75 DB MINIMUM	85 DB TYPICAL

1.2.1-4 NADC ANALOG INPUT

THE ANALOG VOLTAGES OF THE THERMOCOUPLES ARE CONVERTED TO DIGITAL DATA ON THE NADC CARD.

THE CPU ADDRESSES THE NADC CARD ON THE ADDRESS BUS (THE SYSTEM HAS CAPABILITY FOR FOUR NADC CARDS). THE ADDRESS IS DECODED TO SELECT ONE OF THE TWO INPUT/OUTPUT (8255) CHIPS ON THE NADC CARD. THE DECODED ADDRESS ALSO SELECTS ONE OF THE THREE PORTS A, B OR C OF THE I/O CHIP.

THE I/O PORTS ARE INTERFACED WITH THE CPU BI-DIRECTIONAL DATA BUS.

I/O #1	PORT A	THERMOCOUPLE DATA (LOWER BYTE)
	PORT B	THERMOCOUPLE DATA (UPPER BYTE)
	PORT C	THERMOCOUPLE POINT ADDRESS (ON THE NTSA CARDS)
I/O #2	PORT A	CONTROL FUNCTION DATA
	PORT B	NOT USED
	PORT C	UPPER FOUR BITS NOT USED
		LOWER FOUR BITS A/D CONVERSION STATUS INFORMATION

THE CPU SELECTS ONE ANALOG POINT ON THE NTSA CARD AND THIS VOLTAGE IS INPUT TO THE NADC CARD. THE A/D CONVERTER STARTS THE CONVERSION UPON RECEIVING THE CONTROL SIGNAL,

CPU'S CONTROL BUS SIGNALS AND CLOCK (Ø2) ARE GATED ON THE NADC TO GENERATE CONTROL AND TIMING SIGNALS.

ANALOG INPUT RANGES : ± 10 VOWS (FULL SCALE)

OUTPUT 12 BIT
LOGIC "0" LEVEL = .25 ± .2 VDC
LOGIC "1" LEVEL = 3.9 ± 1.2 VDC

SCALE FACTOR : + 10 VDC = 2047 COUNTS
0V = 0 COUNTS

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ELEMENTARY DIAGRAM COMBUSTION MONITOR CARD APPLICATION NOTES
GEK 65322
CONT. ON SH. 11 SH. NO. 10

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1.2.2 CENTRAL PROCESSING UNIT

THE HMPA MICROPROCESSOR BOARD IS THE HEART OF THE COMBUSTION MONITOR SYSTEM. A PROGRAM STORED IN PROGRAMMABLE MEMORY DIRECTS THE FUNCTIONS OF THE HARDWARE. THE PROCESSOR COMMUNICATES DIRECTLY WITH THE OPERATOR DISPLAY MODULE, THE ISOLATED LOGIC INPUT CARD, THE RELAY OUTPUT CARD AND THE ANALOG INPUT SCANNING SYSTEM.

1.2.2.1 HMPA - MICROPROCESSOR

MAJOR COMPONENTS ON THIS CARD ARE :

1. 8 BIT PARALLEL DATA MICROPROCESSOR (INTEL 8080).
2. 4096 BYTES OF ULTRA VIOLET ERASABLE MEMORY PROM, (INTEL **2708)**
3. 1024 BYTES OF READ/WRITE MEMORY RAM. (INTEL 2111)
4. SYSTEMS TIMING CIRCUITRY AND'SYSTEMS CONTROL (INTEL **8238)**
5. I/O CHIPS (INTEL 8255).
6. EIGHT LEVEL PRIORITY INTERRUPT. (INTEL 8259)
7. PARITY GENERATOR
8. USART (INTEL 8251)
9. INTERFACE CIRCUITRY BETWEEN THE MICROPROCESSOR AND THE MEMORY AND I/O CHIPS,

THE SYSTEMS MAIN PROGRAM IS STORED IN THE 4K ULTRA VIOLET ERASABLE MEMORY (PROM). THE 1K RAM MEMORY IS USED FOR TEMPORARY STORAGE OF DATA FROM THE MICROPROCESSOR OR ANY OF THE I/O DEVICES. THE TURBINE THERMOCOUPLE READINGS ARE STORED IN THIS MEMORY,

THE CLOCK GENERATOR/DRIVER CIRCUIT CONSISTS OF A CRYSTAL OSCILLATOR, A FOUR BIT COUNTER AND GATING CIRCUITRY. IWO OUT OF PHASE CLOCK PULSES ARE GENERATED FOR THE MICROPROCESSOR TIMING REFERENCE.

DATA IS TRANSMITTED BETWEEN THE MICROPROCESSOR MEMORY AND THE I/O DEVICES ON A BI-DIRECTIONAL 8 BIT TIME SHARE BUS. INFORMATION IS GATED ON OR OFF THE DATA BUS BY SELECTING OR DESELECTING (TRI-STATING) MEMORY AND I/O DEVICES, (TRI-STATE IS THE ABILITY TO FORCE THE OUTPUT DRIVERS INTO A HIGH IMPEDANCE MODE AS IF IT WERE REMOVED FROM THE CIRCUIT).

A 16 BIT UNIDIRECTIONAL ADDRESS BUS IS USED FOR ADDRESSING MEMORY OR I/O PORTS.

A SET OF UNIDIRECTIONAL CONTROL LINES EMANATE FROM THE MICROPROCESSOR, THESE INDICATE THE TYPE OF ACTIVITY IN PROCESS LIKE MEMORY READ OR MEMORY WRITE ETC.

TWO I/O CHIPS ARE MOUNTED ON TH'E HMPA CARD TO INTERFACE WITH OPERATOR DISPLAY PANEL INPUTS (THUMBWHEEL SWITCHES, RESET SWITCH) AND OUTPUTS (STATUS LEDS, NUMERIC DISPLAY). DISPLAY DATA IS TRANSFERRED TO OR FROM THE PROCESSOR AND THE I/O PORTS OF THE TWO I/O CHIPS.

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I/O PORT ASSIGNMENTS FOR THE DISPLAY BOARD ARE:

I/O CHIP #1 PORT A AND B = FOUR DIGIT NUMERIC DISPLAY.
I/O CHIP #1 PORT C = NOT USED.
I/O CHIP #2 PORT A = TWO THUMBWHEEL SWITCHES
I/O CHIP #2 PORT B = LED STATUS DISPLAY. LEDS ON THE HMPA ALSO DISPLAY THE CONDITION.

I/O #2 PORT C LOWER FOUR BITS = BLANKING AND DECIMAL POINTS, (BITS O-BIT 3)
I/O #2 PORT C (BIT 6) = RESET' PUSH BUTTON (IN),

DATA COMMUNICATION IS HANDLED BY THE USART (INTEL #8251) ON THE HMPA CARD. THE USART ACCEPTS PARALLEL DATA FROM THE CPU AND TRANSMITS SERIAL STREAM OF DATA, IT ACCEPTS SERIAL DATA INPUT AND SENDS PARALLEL DATA TO THE PROCESSOR,

AN EIGHT LEVEL PRIORITY INTERRUPT CHIP (INTEL 8259) IS USED TO DETERMINE THE HIGHEST PRIORITY INTERRUPT 'REQUESTED AND ISSUE AN INTERRUPT REQUEST TO THE PROCESSOR, MAXIMUM UTILIZATION OF THE MICROPROCESSOR CAPACITY IS ACHIEVED BY THE INTERRUPT CHIP WHEN A TERMINET® TERMINAL OR AN RS232 COMPATIBLE TELETYPE® TERMINAL IS BEING USED. THE MICROPROCESSOR CAN RETURN TO SOME OTHER TASK AND WAIT FOR AN INTERRUPT FROM THE TERMINET® OR TELETYPE® TERMINAL.
A PARITY GENERATOR CIRCUIT IS ALSO LOCATED ON THE HMPA CARD, GOOD PARITY IS INDICATED BY THE YELLOW LED BEING LIT. THIS CIRCUIT HAS BEEN DISABLED IN THE COMBUSTION MONITOR SYSTEM AND THE YELLOW LED STAYS LIT,

THE INTEL 8080 CHIP REQUIRES +12VDC, +5VDC AND -5VDC. THE +12VDC IS TAPPED OFF THE #15VDC POWER SUPPLY BUS. THE -5VDC IS TAPPED OFF THE -15VDC POWER SUPPLY BUS, +5VDC IS AVAILABLE ON THE SEM BUS, REFER TO INTEL MANUALS FOR HARDWARE DETAILS.

1.2.3 INPUT/OUTPUT I NTERFACE W ITH TURB I NE CONTROL PANEL

1.2.3-I HLIC = ISOLATED INPUT CARD

THE DS3800HLIC LOGIC INPUT CARD PROVIDES FOR EIGHT LOGIC INPUTS: THESE INPUTS ARE STANDARD TTL 5.25 VOLTS MAXIMUM AND ARE AVAILABLE EITHER AT THE SEM RIBBON CABLE PLUG JK OR TERMINAL BOARD TBA ON THE INS IDE ENCLOSURE WALL. ISOLATION IS OBTAINED BY RELAY I NPUT CONTACTS, THE PROCESSOR COMMUNI - CAT ES DIRECTLY WITH THE HLIC THRU THE DIRECTOMATIC PLUS R-BUS (SEE GEK 61680 FOR DEFINITION OF F AND R BUSSES).

A PARITY GENERATOR CIRCUIT IS DESIGNED ON THE BOARD. GOOD PARITY IS INDICATED BY THE YELLOW LED BEING LIT. THIS CIRCUIT IS DISABLED IN THE COMBUSTION MON- ITOR APPLICATION AND THE LED STAYS ON.

1.2.3-Z HRRA = RELAY OUTPUT

THE DS3800 HRRA RELAY OUTPUT CARD CONTAINS EIGHT DOUBLE POLE RELAYS THAT OUTPUT THE ALARM OR TRIP SIGNALS TO THE TURBINE CONTROL PANEL. FOUR RELAYS HAVE IWO NORMALLY CLOSED CONTACTS EACH AND FOUR RELAYS HAVE TWO NORMALLY OPEN CONTACTS EACH, ONE CONTACT OF EACH RELAY IS W I RED TO THE SEM RIBBON CABLE PLUG JK FOR LOW LEVEL S IGNAL W I R ING AND ONE CONTACT IS W IRED TO THE TERMI NAL BOARD TBB ON THE INS IDE ENCLOSURE WALL FOR HIGH LEVEL (S IGNAL WIRING (125VDC). EACH RELAY is ADDRESSABLE THRU THE DIRECTOMATIC PLUS R-BUS.

THE RELAY CARDS ARE OPERATED ON P28V, CONTACT RATINGS ARE

D.C. RESISTIVE = 2 AMPS AT 28 VOLTS
D.C. INDUCTIVE = 0.5 AMPS AT 28 VOLTS, 200 LEFT
125 D.C. INDUCTIVE OR RESISTIVE 150 MILLI AMPS L/R 5 MSEC.
A.C. RESISTIVE = 0.5 AMPS AT 115 VOLTS
A.C. INSUCTIVE = 0.5 AMPS CONT., 1.5 AMPS INRUSH FOR 115 VOLTS AC L/R

A PARITY CHECK CIRCUIT IS DESIGNED ON THE BOARD, GOOD PARITY IS INDICATED BY THE YELLOW LED BEING LIT. IN THE COMBUSTION MONITOR APPLICATION THE PARITY CIRCUIT HAS BEEN DISABLED AND THE YELLOW LED STAYS ON,

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1.2.4 OPERATOR DISPLAY MODULE

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THE DISPLAY MOD'JLE (DS3820DP1) CONSISTS OF A DISPLAY PWB (DS3800HDDB), A TWO DIGIT THUMBWHEEL SW ITCH AND A PUSHBUTTON. IT IS MOUNTED IN THE FRONT DOOR OF THE COMBUSTION MONITOR ENCLOSURE, NINE LED'S (LIGHT EMITTING DIODES) INDICATE THE "TURBINE STATUS" INFORMATION.

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1.2.4-1 HDDB - D IS PLAY I NTERFACE CARD

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THE DS3800 HDDB PROVIDES. INTERFACE To DISPLAY DATA FROM THE CPU, IT INPUTS COMMANDS, FROM THE THUMBWHEEL SWITCHES AND PUSHBUTTON MOUNTED ON THE DISPLAY PANEL. THE HDDB CARD IS MOUNTED ON THE DOOR. RIBBON CABLES MAKE CONNECTIONS ON THE SEM AND THE HMPA CARD.

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THE SYSTEMS STATUS IS DISPLAYED ON LED'S ON THE FRONT PANEL. TURBINE AND MON-ITOR STATUS DISPLAY DATA IS OUTPUT FROM I/O CHIP #2 (PORT B) OF THE HMPA CARD. "O TRUE" OUTPUTS FROM THE HMPA CARD W I LL ENERG I ZE THE LEDS. THE 'TURBINE STATUS' LEDS WHEN ENERG IZED I ND I CATE THE FOLLOW i NG CONDITIONS.

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21 THERMOCOUPLE FAILURE
COMBUSTION ALARM
NORMAL SHUTDOWN
23 TURBINE TRIP
SPARE (WILL INDICATE "WATER SYSTEM ON" FOR TURBINES WITH WATER
25 INJECTION)

27 THE " MON ITOR STATUS" LED'S WHEN ENERGI ZED, I ND I CATE THE FOLLOW I NG COND I T IONS

29 POWER ON	(LED IS ON WHEN POWER SUPPLY IS ON)
ENABLED	
NORMAL	
31 STALLED	(NORMALLY OFF UNTIL A STALL S IGNAL IS GENERATED BY THE HMPA CARD)

33

A FOUR DIGIT HEXADECIMAL DISPLAY IS AVAILABLE AND LABELLED "POINT VALUE". 35 I/O CHIP #1 (PORTS A AND B) OF THE HMPA CARD OUTPUT "O TRUE" SIGNALS TO DRIVE THE DISPLAY CHIPS.

37

THE THUMBWHEEL SW ITCH IS LABELLED "POINT SELECT". IT PROV I DES 100 DATA PO I NTS AS INDICATED ON THE THUMBWHEEL SELECTOR SCHEDULE ON THE DOOR, THE OUTPUTS ARE 39 PULLED UP TO 5 VOLTS, BUT W ITH THE THUMBWHEEL COMMON TIED TO DCOM, A *TRUE" OUTPUT IS O. DATA FROM THE THUMBWHEEL SW ITCH IS FED I NTO I/O CH IP #2 (PORT A) OF THE 41 HMPA CARD.

43

A MOMENTARY CONTACT RESET PUSH BUTTON IS LABELLED "RESET", IT IS USED IN COMBINATION WITH THE POINT SELECT THUMBWHEEL SW ITCHES FOR SYSTEM RESET AND 45 TEST MODES. -OUTPUT IS LOGIC 0 NORMALLY, LOGIC 1 WHEN THE PUSH BUTTON IS PRESSED,

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REV. 1 1/25/79 AG	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM COMBUSTION MONITOR CARD APPLICATION NOTES
REV. 2	ISSUED 9-15-78			FIRST MADE FOR		GEK65322
REV. 3	MADE BY AZRA- EHSAN			I C. NO.		CONT. ON SH. 15 SH. NO. 14

A B C D E F G H J K L

01

03

DS3820PS3 POWER SUPPLY

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THE POWER SUPPLY CAN OPERATE ON AC OR DC INPUT VOLTAGES. THE DC INPUT VOLTAGE RANGE IS FROM 90 VDC TO 140 VDC, WITH MOMENTARY DIPS TO 70 VDC. THE

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AC INPUT VOLTAGE -RANGE IS FROM 80 VRMS TO 132 VRMS, 47-63 HZ. THE INPUT CURRENT IS 4.7 AMPS RMS MAXIMUM OR 5.5 AMPS D.C. MAXIMUM (100 A 1 CYCLE INRUSH),

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IT PROVIDES PARALLEL OUTPUTS OF:

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+5 ± .15 VDC AT 22.5 AMPS, 50MV P.P RIPPLE MAX
+15 ± .15 VDC AT 2 AMPS, 50MV P.P RIPPLE MAX
-15 ± .15 VDC AT 2 AMPS, 50MV P.P RIPPLE MAX
+28 ± .5 VDC AT 1 AMP, 100V P.P RIPPLE MAX

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TWO TERMINAL BOARDS TB1 AND TB2 ARE LOCATED ON THE POWER SUPPLY FOR POWER CONNECTIONS AND JUMPERS.

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POWER CONNECTIONS ARE MADE AT TB1-1 (HIGH) AND TB1-2 (COMMON).

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CHASS IS GROUND IS CONNECTED AT TB1-4.
JUMPERS ON POWER SUPPLY TERMINALS ARE IN THE FOLLOWING CONFIGURATION:

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TB1 (4) TO TB1 (5)
TB1 (5) TO TB1 (6)
TB2 (1) TO TB2 (2)
TB2 (7) TO TB2 (8) FOR DC OPERATION, TB2 (7) TO TB2 (6)
FOR AC OPERATION:

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REGULATED OUTPUT DC VOLTAGES FROM THE POWER SUPPLY ARE AVAILABLE ON TWO

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PARALLEL OUTPUT CONNECTORS J1 AND J2 LOCATED ON THE RIGHT SIDE OF THE POWER SUPPLY, THE CONNECTOR J1 IS DIFFERENT FROM J2 IN THAT J1 HAS R5H AND R5L

31

REMOTE SENSE LINES USED TO REGULATE THE +5VDC AT THE LOAD, SEM CONNECTOR JA IS CONNECTED TO CONNECTOR J1 OF THE POWER SUPPLY, CONNECTOR J2 MAY BE USED FOR ANY AUXILIARY EQUIPMENT.

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THE MONITOR POWER IS ENERGIZED BY THE CIRCUIT BREAKER CB1. THE OUTPUT VOLTAGES

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ARE LOCATED ON J1 AND J2. OUTPUT CURRENT AND VOLTAGE CAN BE MONITORED VIA TEST POINTS LOCATED ON THE CONTROL CARD (DS3800NPSC). THE POWER SUPPLY HAS OVERVOLTAGE

37

PROTECTION AND SHORT CIRCUIT CURRENT LIMITING ON EACH OUTPUT VOLTAGE. AN MOV (METAL OXIDE VARISTOR) SURGE SUPPRESSOR PROTECTS THE INPUT AGAINST LINE TRANSIENTS

39

OF SHORT DURATION. A BALUN CIRCUIT CONSISTING OF A SPLIT REACTOR AND TWO CAPACITORS IS USED AS EMI FILTER TO PREVENT SWITCHING SPIKES FROM BEING DEVELOPED ACROSS THE INPUT POWER LINE.

41

REMOTE SENSING LINES ARE AVAILABLE TO ENABLE REGULATION AT A REMOTELY CONNECTED

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LOAD, SENSOR OUTPUTS ARE UTILIZED TO PROVIDE A LOGIC SIGNAL TO EXTERNALLY CONNECTED ELECTRONICS IF ANYONE OF THE INPUT VOLTAGES FALLS BELOW A SET PERCENTAGE OF NORMAL VOLTS (5% FOR P5, P15 AND N15, 20% FOR p28).

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A STATUS LIGHT IS LOCATED ON THE CONTROL CARD AND MONITORS POWER SUPPLY STATUS.

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WHEN IT IS "ON" ALL POWER SUPPLY OUTPUTS ARE NORMAL.

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IF THE POWER SUPPLY IS FUNCTIONING PROPERLY, IT SHOULD ONLY BE NECESSARY TO TRIM THE OUTPUT VOLTAGES AT THE EXTERNAL LOADS. THE FOLLOWING ADJUSTMENTS CAN BE MADE WHILE MONITORING THE LOAD VOLTAGE ;

P5 ± .05 VDC	R1 (ON POWER SUPPLY CONTROL CARD)
P15 ± .05 VDC	R9 (ON POWER SUPPLY CONTROL CARD)
N15 ± .05 VDC	R10 (ON POWER SUPPLY CONTROL CARD)
P28 + .5 VDC	R8 (ON POWER SUPPLY CONTROL CARD)

THE POWER SUPPLY WILL DELIVER FULL OUTPUT FOR AMBIENT TEMPERATURES FROM 0 TO 50° C.

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ELEMENTARY DIAGRAM
COMBUST I ON MONITOR CARD APPL I CAT I ON NOTES

GEK 65322

CONT. ON SH. FL SH. NO. 16